



Answer any three (03) out of the following questions:

[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

$$3 \times 10 = 30$$

CLOs Marks

CLO1 [2]

CLO3 [8]

$$\frac{1}{n} \sum_{i=1}^n x_i$$

$$\frac{1}{n} \sum_{i=1}^n x_i^2$$

Q1. (a)

(b)

(c)

What are the primary types of data (qualitative and quantitative)? Explain with examples.

Find the arithmetic mean, geometric mean, harmonic mean from the following frequency distribution of weekly wages with a class interval of 50 taka each.

Class Interval of Wages (Taka)	Number of Workers (f)
10-15	5
15-20	8
20-25	12
25-30	15
30-35	7
35-40	3
Total	50

Q2. (a)

(b)

(c)

Find the relation between arithmetic mean, geometric mean and harmonic mean. Why is it used? Provide one example.

Find the median from the frequency distribution of weekly working hours recorded for 40 employees in a factory as shown in the table below:

Class Interval of Working Hours	Number of Employees (f)
30-35	5
35-40	7
40-45	8
45-50	10
50-55	6
55-60	4
Total	40

$$\frac{1}{n} \sum_{i=1}^n x_i$$

Q3. (a)

(b)

(c)

What are the four levels of measurement in statistics? Define each with an example.

Define independent and dependent events in probability. Explain their difference.

A bag contains 12 identical balls, of which 4 are red and the rest are blue. Two balls are drawn at random from the bag. What is the probability that:

1. Both balls are red?

2. One ball is red and the other is blue?

Q4. (a)

(b)

(c)

Define sample space with example.

CLO1 [1]



- (b) Find the mean deviation from the mean and the standard deviation of the observations $a, a + d, a + 2d, \dots, a + 2nd$. Prove that the standard deviation is greater than the mean deviation.
- (c) Three groups of students contain the following numbers of boys and girls:

- Group 1: 2 girls and 3 boys
- Group 2: 4 girls and 2 boys
- Group 3: 1 girl and 5 boys

One student is selected at random from each group. Find the probability that the three selected students consist of exactly 2 boys and 1 girl.

Q5.

- (a) Define standard deviation.
- (b) Using the frequency distribution given in the following table, calculate:
1. The mean deviation from the mean.
 2. The mean deviation from the median.
- Verify your calculations with proper steps and explanations.

Class Interval (Variable)	Frequency (fi)
0-5	2
5-10	5
10-15	7
15-20	13
20-25	21
25-30	16
30-35	8
Total	72

- (c) Prove that the standard deviation of the first n natural numbers is given by the

$$\text{Formula: } \sqrt{\frac{(n^2 - 1)}{12}}$$



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 213: Digital Electronics

Batch: 25th
Date: 04/01/2024

Mid-Term Examination

Semester: Fall 2024
Time: 1 Hour 30 Minutes
Total Marks : 30

Answer any **five** questions:

$5 \times 6 = 30$

1. (a) Distinguish between digital and analog signal representations. 3
(b) What are the advantages of digital techniques? 3
2. (a) Draw a flowchart for repeated-division method of decimal-to-octal conversion of integers. 3
(b) Illustrate the functional diagram of a digital computer. 3
3. (a) Prove the universality of NAND and NOR gates. 3
(b) Demonstrate Demorgan's theorems and prove them. 3
4. (a) Simplify the expression $Z = ABC + \overline{AB}(\overline{AC})$ and draw circuits for both expressions. 3
(b) Draw a circuit diagram for the equation $X = \overline{ABC} + \overline{DBC} + (\overline{ABD} + \overline{ABC})$. 3
5. (a) What is Flip-Flop? Summarize NAND Latch. 3
(b) Draw and explain the operation of clocked J-K Flip-Flop. 3
6. (a) Sketch and explain parallel data transfer between two registers. 3
(b) Briefly explain arithmetic overflow, negation and 2^s complement. 3
7. (a) Describe BCD and ASCII codes. 3
(b) Represent $(1358)_{10}$ and $(2653)_{10}$ in BCD and then perform BCD addition. Check the result by converting back to decimal. 3



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Bachelor of Science in CSE

BAN211: History of Emergence of Bangladesh

Mid-Term Examination

Fall 2024

25th Batch

01 January 2025

3rd Semester

Time: 1 hour 30 minutes

Total Marks: 30

Part I: Short question (Answer any 4)

- ✓ 1. List the key activities of the Dutch in India. 05
- ✓ 2. Explain the reasons behind Mir Qasim's defeat in the Battle of Buxar. 05
- ✓ 3. Explain why the Sepoy Mutiny failed to succeed. 05
- ✓ 4. Evaluate the results of the Battle of Plassey and their impact. 05
5. Assess the aim of the Faraizi movement. 05

Part II: Broad question (Answer any one)

6. Analyze the reasons behind the British arrival in Bengal and India and examine how they established their dominance in the region. 10
- ✓ 7. Analyze the Permanent Settlement, its purpose, and examine its negative impacts on Indian society and economy. 10



HAMDARD UNIVERSITY BANGLADESH

FACULTY OF SCIENCE, ENGINEERING AND TECHNOLOGY

ACT 215: Financial and Managerial Accounting

Semester: Fall 2024

08/01/2025

Mid-term Examination 2024

Time: 90 minutes

Total Marks: 10x3=30

Instructions:

- All parts of each question must be answered consecutively.
- The figures in the right margin indicate full marks.
- Answers to questions should be neat and clean.
- Answer any **three (3)** of the following questions.

- 1 Tony Anderson opened a veterinary business in New York, on August 1, 2024. On August 31, the balance sheet showed Cash Tk.36,000, Accounts Receivable Tk.7,700, Supplies Tk.1,200, Office Equipment Tk.23,000, Accounts Payable Tk.20,600, and M. Gonzalez, Capital Tk. 47,300. During September the following transactions occurred.
- September 1 Paid Tk. 17,900 cash on accounts payable..
- 5 Collected 6,300 of accounts receivable..
- 7 Purchased additional office equipment for 11,100, paying 9,200 in cash and the balance on account..
- 10 Earned revenue of Tk.37,000, of which Tk.24,500 is paid in cash and the balance is due in October.
- 15 Withdrew Tk.5,000 cash for personal use..
- 17 Paid salaries Tk.10,700, rent for September Tk.1,800, and advertising expense Tk. 1,600.
- 20 Incurred utilities expense for month on account Tk.1,170.
- 25 Received Tk.13,600 from Capital Bank—money borrowed on a note payable.
- 28 Paid Tk.1,550 for insurance.
- 30 Received Tk.6,000 in cash from customers who have previously been billed on September 10.

Requirement: i) **Demonstrate** a tabular analysis of above transaction. The column headings should be as follows: Cash, A/R, Supplies, Office equipment, A/P, N/P, Tony's capital.

ii) **Develop** an Income Statement and Balance Sheet for the month of September, 2024.

08

02

96450
24550

57050

4100

34100

CLO2

ex - en In - on
or



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- 2 Nandan Park was started on June 1 by Nandan. The following selected events and transactions occurred during June. 10 CLO3
- June. 1 Nandan invested Tk.2,80,800 cash to start the business.
June. 4 Purchased land costing Tk.1,15,000 for cash.
June. 8 Incurred advertising expense of Tk.13,550 on account.
June. 11 Paid salaries to employees Tk.12,500.
June. 12 Hired park manager at a salary of Tk.34,000 per month, effective July 1.
June. 13 Paid Tk.4,800 cash for a one-year insurance policy.
June. 17 Withdrew 11,000 cash for personal use.
June. 20 Received Tk.8,700 in cash for admission fees.
June. 25 Sold 500 coupon books for Tk.35.50 each. Each book contains 10 coupons that entitle the holder to one admission to the park.
June. 30 Received Tk.9,700 in cash admission fees.
June. 30 Paid Tk.13,550 on balance owed for advertising incurred on June 8.
Construct Journal for each transaction.
- 3 a) Define accounting as per AICPA. 02 CLO1
b) Identify the users of accounting information and how they use this information. 04
c) Discuss the different merit and limitation of accounting 04
- 4 a) Explain the different function of management accounting. 03 CLO1
b) Show the different nature of management accounting. 04
c) Analyze the tools and technique of management accounting 04
- 5 Suripure Miniature Golf and Driving Range was opened on July 1 by Tony Anderson. The following selected events and transactions occurred during July: 02 CLO3
- July 1; Invested Tk.1,25,000 cash in the business. 02
July 3; Purchased Lee's Golf Land for Tk.78,000 cash. The price consists of land Tk.53,000, Buildings Tk.15,000 and equipment Tk.10,000 02
July 5; Advertised the opening of the driving range and miniature golf course, paying advertising expenses of Tk.4,600. 02
July 6; Paid Cash Tk.5,000 for a one-year insurance policy. 02
July 10; Purchased golf clubs and other equipment for Tk.15,600 from Parton company payable in 30 days.
July 18; Received Tk.10,800 in cash for golf fees earned.
July 19; Sold 1200 Coupon books for Tk.15 each.
July 25; Withdrew Tk.5,000 cash for personal use.
July 30; Paid Parton Company in full.
July 30; Received Tk.11,000 cash for fees earned.
Construct Journal for each transaction.



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Department of CSE

CSE 215: Data Structure

Mid-Term Examination

Fall 2024

Batch: 25th

Date: 06/01/2025

Mid Term Examination

Semester: 3rd Semester

Time: 1 Hour 30 Minutes

Total Marks: 30

Answer any **five** questions from the following:

5x6= 30

	CLOs	Marks
1. (a) Outline the differences between primitive and non-primitive data structure.	CLO1	2
(b) Explain the basic operations of data structures with example.	CLO5	4
2. (a) Why linked list is more efficient than array?	CLO1	2
(b) Demonstrate the formal algorithm for traversing a singly linked list.	CLO1	4
3. (a) Define overflow and underflow in the context of linked list.	CLO1	3
(b) A String with different characters say "MMPPQRRSTTTUUUUUV". Find the Huffman code for this string.	CLO2	3
4. (a) Utilize Insertion Sort to sort the array [12, 11, 13, 5, 6] and explain each step of the algorithm.	CLO2	3
(b) Apply Quick Sort on the array [10, 80, 30, 90, 40, 40, 50, 70] using last element as the pivot. Show all partition steps.	CLO2	3
5. (a) Why Binary Search is more efficient than Linear Search for sorted datasets? Compare their time complexities.	CLO4	2
(b) Apply Binary Search to locate the element 15 in the sorted array [2, 5, 7, 11, 15, 19]. Show the process step by step.	CLO2	4
6. (a) Change the following infix expression to prefix and postfix expression: $(A+B)*C-D/E$.	CLO4	2
(b) Change the following expression to postfix using Stack: $A+B*(C^{\wedge}D-E)/F+G$.	CLO4	4

CLO4 2
CLO2 4

7. (a) Analyze the drawbacks of linear queue data structures.
(b) Consider an empty Circular Queue with a capacity of 6.
- Insert 30.
 - Insert 50.
 - Insert 70.
 - Delete Front.
 - Insert 90.
 - Delete Front.
 - Insert 130.
 - Delete Front.
- Determine the state of the Circular Queue after each of the above operation.